

Name: AAVISHKAR SINGH AAVISHKAR SINGH | Reg. No: RA2511003011024

Unit 1 : Session 4 : SLO 2 | Implementation of Structures Using Pointers Fill in the Blanks

I. Fill in the Blanks

1. * (asterisk) a structure pointer is declared using the * operator: struct S *ptr;
2. -> (arrow) the arrow operator is used to access members via a pointer.
3. the address a structure pointer stores the address of a structure variable.
4. &var ptr = &var; assigns the address.
5. direct (efficient member) pointers enable direct access to the structure's members without copying.
6. allocated (initialized) memory for the pointed structure must be allocated before use.
7. in-place (permanent) passing a pointer to a function lets it modify the original structure.
8. ptr->member (*ptr).member is the same as ptr->member.
9. structure (data) pointers reduce structure copying overhead.
10. malloc dynamic allocation: ptr = malloc(sizeof(struct S));
11. size (amount of memory) sizeof(struct) ensures the correct allocation size.
12. freed unused structure pointers' memory should be freed.
13. dynamic pointers enable dynamic structures like linked lists.
14. dangling pointers (undefined behavior) always initialize to NULL or a valid address.
15. member access (the dot) the arrow operator combines dereference and member access.